

Monopoly

ECON410 Intermediate Microeconomics

Monopoly Profit Maximization

- Like all firms, monopolies maximize profit by producing the quantity where marginal revenue (MR) equals marginal cost (MC).

$$MR(Q^*) = MC(Q^*)$$

- Unlike perfectly competitive firms, a monopolist's profit maximization problem is the optimal trade-off between P & Q.
 - For any positive Q, marginal revenue is less than the price the monopolist can charge to sell that quantity.

$$MR(Q) = \frac{dR(Q)}{dQ} = \frac{dp(Q)Q}{dQ} = p(Q)\frac{dQ}{dQ} + \frac{dp(Q)}{dQ}Q = p(Q) + \frac{dp(Q)}{dQ}Q$$

Example:

$$P = 240 - Q$$

What is its MR?

In general:

$$P = a - bQ$$

$$MR = a - 2bQ$$

$$MR = P(Q) + \frac{dP(Q)}{dQ} \cdot Q$$

	P	Q	$P \cdot Q =$ TR	$\frac{TR}{Q} =$ AR	$\frac{\Delta TR}{\Delta Q} =$ MR
	100	1	100	100	
	98	2	196	98	96
	96	3	288	96	92
	94	4	376	94	88
	92	5	460	92	84

Example

- $P = 240 - Q$
- $TC = Q^2 + 12$
- Calculate the monopolist's profit π

Example

- $P = 240 - Q$
- $TC = Q^2 + 12$
- Calculate the monopolist's profit π
- $\pi = 60$

Monopoly Example

- The monopolist's profit maximizing choice of output is found where $MR=MC$.
- At the profit-maximizing output, set p according to inverse demand.
- If $TC = Q^2 + 12$, the profit maximizing monopolist will produce 6 units and charge \$18 per unit.

